(2). In a certain factory turning out razor blades, there is a small chance of 0.002 for any blade to be defective. The blades are supplied in packets of 10, use Possion distribution to calculate the approximate number of packets containing no defective, one defective and two defective blades respectively in a consignment of 10,000 packets.

▲ Given that $n = 10$ and $p = 0.002$

$\therefore m = np = 10 \times 0.002 = 0.02$

$$P(x) = \frac{e^{-m} m^x}{x!} = \frac{e^{-0.02} \times [0.02]^x}{x!}$$

Let x denote the no. of defective blades.

$$P[x=0] = \frac{e^{-0.02} \times [0.02]^0}{0!} = e^{-0.02} = 0.9802$$

$\therefore$ Probability of no defective blade is $= 0.9802$.

$\therefore$ no. of packets containing no defective blade is $= 10000 \times 0.9802 = 9802$

$$P[x=1] = \frac{e^{-0.02}[0.02]}{1!}$$

$$= e^{-0.02} \times [0.02] = 0.02 \times 0.9802 = 0.019604$$

$\therefore$ The no. of packets containing one defective blade

$= 10,000 \times 0.019604 = 196.04 \approx 196$

$$P[x=2] = \frac{e^{-0.02}[0.02]^2}{2!} = \frac{[0.02]^2}{2} \times 0.9802$$

Finally, the no. of packets containing two defective blades $= 10,000 \times \frac{(0.02)^2}{2} \times 0.9802 = 2$ approximately.

(3) ... distribution to the set of

⑤. If a random variable has a Poisson distribution such that $P(1) = P(2)$ [i.e $P[x=1] = P[x=2]$]. Find

(i) mean of the distribution (ii) $P(4)$.

▲ We have $P(x) = \frac{e^{-m} m^x}{x!}$

Given that $P(1) = P(2) \Rightarrow \frac{e^{-m} m^1}{1!} = \frac{e^{-m} m^2}{2!}$

$\Rightarrow m^2 = 2m \Rightarrow \boxed{m=2}$

(i) Mean of the distribution : $m=2$

(ii) $P(4) = \frac{e^{-m} m^4}{4!}$ Since $m=2$

$= \frac{e^{-2} 2^4}{4!} = \frac{2}{3} e^{-2}$

⑥. X is a Poisson variable and it is found that the Probability that $X=2$ is two-thirds of the Probability that $X=1$. Find the Probability that $X=0$ and the Probability that $X=3$. What is the Probability that X exceeds 3?

Given that $P(2) = \frac{2}{3} P(1)$

we have $P(x) = \frac{e^{-m} m^x}{x!}$

$P(2) = \frac{2}{3} P(1) \Rightarrow \frac{e^{-m} m^2}{2!} = \frac{2}{3} \frac{e^{-m} m^1}{1!}$

$\Rightarrow m^2 = \frac{4}{3} m$

$\therefore m = \frac{4}{3}$

$\therefore P(x) = \frac{e^{-(4/3)} (4/3)^x}{x!}$

$P[x=0] = \frac{e^{-4/3} \times (4/3)^0}{0!} = e^{-4/3} = 0.2636$

$P[x=3] = \frac{e^{-4/3} \times (4/3)^3}{3!} = \frac{64}{27 \times 6} \times e^{-4/3} = 0.1041$

$P[x>3] = 1 - P[x \leq 3]$

$= 1 - \{ P[x=0] + P(x=1) + P[x=2] + P[x=3] \}$

$= 1 - \{ 0.2636 + \frac{e^{-4/3} (\frac{4}{3})^1}{1!} + \frac{e^{-4/3} (\frac{4}{3})^2}{2!} + 0.1041 \}$

(∵ From the above we have $P[x=0] = 0.2636$, $P[x=3] = 0.1041$)

$= 0.1506$

(7). A certain screw making machine produces an average of 2 defective screws out of 100, and packs them in boxes of 500. Find the probability that a box contains 15 defective screws.

▲ Given that $p = \frac{2}{100} = 0.02$ and $n = 500$

Mean $m = np = 500 \times 0.02 = 10$

We have $P(x) = \frac{e^{-m} m^x}{x!}$ | Let x denotes the no. of defective screws

$$P(x) = \frac{e^{-10} \times 10^x}{x!}$$

$$\therefore P(x=15) = \frac{e^{-10} \times 10^{15}}{15!} = 0.035.$$

(8). A manufacturer knows that the condensers he makes contain on the average 1% defective. He packs them in boxes of 100. What is the probability that a box picked at random will contain 3 or more faulty condensers?

▲ Given that $p = \frac{1}{100} = 0.01$ and $n = 100$

Let x denotes the no. of defective condensers.

$\therefore$ Mean $= m = np = 100 \times 0.01 = 1$

&

$$P(x) = \frac{e^{-m} m^x}{x!} = \frac{e^{-1} 1^x}{x!} = \frac{1}{e}\left[\frac{1}{x!}\right]$$

The probability that a box contains 3 or more defective condensers is $= P[x \geq 3]$

$$= 1 - P[x < 3]$$

$$= 1 - \{P[x=0] + P[x=1] + P[x=2]\}$$

$$= 1 - \left\{\frac{1}{e}\frac{1}{0!} + \frac{1}{e}\frac{1}{1!} + \frac{1}{e}\frac{1}{2!}\right]$$

$$= 1 - \frac{1}{e}\left[1 + 1 + \frac{1}{2}\right]$$

$$= 1 - \frac{5}{2e} = 0.08$$

(9). A car-hire firm has two cars which it hires out day by day. The number of demands for a car on each day is distributed as a Poisson distribution with mean 1.5. Calculate the proportion of days (i) on which there is no demand (ii) on which demand is refused.

Note: $\boxed{e^{-1.5} = 0.2231}$

▶ Given that mean $= m = 1.5$

We have $P(x) = \frac{e^{-m} m^x}{x!} = \frac{e^{-1.5} (1.5)^x}{x!}$

Here, x denotes the demand for a car on a day.

(i) Then the probability on which there is no demand is $= P[x=0] = \frac{e^{-1.5}(1.5)^0}{0!} = e^{-1.5} = 0.2231$

(ii) Next, the probability that some demand is refused

$= $ Probability that there are more than 2 demands

$= P[x > 2]$

$= 1 - P[x \leq 2]$

$= 1 - \{ P[x=0] + P(x=1) + P(x=2) \}$

$= 1 - \left\{ \frac{e^{-1.5}(1.5)^0}{0!} + \frac{e^{-1.5}(1.5)^1}{1!} + \frac{e^{-1.5}(1.5)^2}{2!} \right\}$

$= 0.1913$

(10). The incidence of occupational disease in an industry is such that the workmen have a 10% chance of suffering from it. What is probability that in a group of 7, five or more will suffer from it?

▶ Given that $p = \frac{10}{100} = 0.1$ and $n = 7$

$\therefore$ Mean $= m = np = 7 \times 0.1 = 0.7$

Let 'x' denote the number of people suffering from disease.

$$\therefore P(x) = \frac{e^{-m} m^x}{x!} = e^{-0.7} \frac{(0.7)^x}{x!}$$

The probability of five or more will suffer disease

$$= P[x \geq 5] = 1 - P[x < 5]$$

$$= 1 - \{ P[x=0] + P[x=1] + P[x=2] + P[x=3] + P[x=4] \}$$

$$= 1 - \left\{ e^{-0.7} \frac{(0.7)^0}{0!} + e^{-0.7} \frac{(0.7)^1}{1!} + e^{-0.7} \frac{(0.7)^2}{2!} + e^{-0.7} \frac{(0.7)^3}{3!} + e^{-0.7} \frac{(0.7)^4}{4!} \right\}$$

$$= 0.0008.$$

(11). The number of accidents in a year to taxi drivers in a city follows a Poisson distribution with mean 3. out of 1000 taxi drivers find approximately the number of the drivers with (i) no accident in a year
(ii) more than 3 accidents in a year.

Given that $m = 3$.

Let 'x' denotes the no. of accidents in a year to taxi drivers.

$$\therefore P(x) = \frac{e^{-m} m^x}{x!} = \frac{e^{-3} 3^x}{x!}$$

$$P(x) = \frac{e^{-3} 3^x}{x!}$$

(i) The probability of no accidents in a year is

$$= P[x=0] = \frac{e^{-3} \times 3^0}{0!} = e^{-3} = 0.05$$

$\therefore$ Number of drivers out of 1000 with no accidents in a year $= 1000 \times 0.05 = 50$.

(ii). The probability of more than 3 accidents in a year is $= P[x > 3] = 1 - P[x \leq 3]$

$$= 1 - \{ P[x=0] + P[x=1] + P[x=2] + P[x=3] \}$$

$$= 1 - \left\{ \frac{e^{-3} \times 3^0}{0!} + \frac{e^{-3} \times 3^1}{1!} + \frac{e^{-3} \times 3^2}{2!} + \frac{e^{-3} \times 3^3}{3!} \right\}$$

$$= 1 - [0.05 + (0.05)\,3 + (0.05)\frac{9}{2} + (0.05) \times \frac{27}{6}\}$$

$$= 0.35$$

$\therefore$ No. of drivers out of 1000 with more than 3 accidents in a year $= 1000 \times 0.35 = 350$.

(12). 2% of the fuses manufactured by a firm are found to be defective. Find the probability that a box containing 200 fuses contains: (i) no defective fuses (ii) 3 or more defective fuses.

▲ Given that $p = \frac{2}{100} = 0.02$ and $n = 200$

$\therefore$ Mean $= m = np = 200 \times 0.02 = 4$.

Let x denote the no. of defective fuses.

$\therefore P(x) = \frac{e^{-m} m^x}{x!} = \frac{e^{-4} \times 4^x}{x!}$ $\boxed{e^{-4} = 0.0183}$

$P(x) = 0.0183 \times \frac{4^x}{x!}$

(i) Probability of no defective fuse $= P[x=0] = 0.0183 \times \frac{4^0}{0!} = 0.0183$.

(ii). Probability of 3 or more defective fuses

$= P[x \geq 3] = 1 - P[x < 3]$

$= 1 - \{P[x=0] + P[x=1] + P[x=2]\}$

$= 1 - \{0.0183 \times \frac{4^0}{0!} + 0.0183 \times \frac{4^1}{1!} + 0.0183 \times \frac{4^2}{2!}\}$

$= 0.7621$

(13). There is a chance that 5% of the pages of a book contain typographical errors. If 100 pages of the book are choosen at random, find the probability that 2 of these pages contain typographical errors, using: (i) Binomial distribution and (ii) Poisson distribution.

▲ (i) Given that using Binomial distribution $p = \frac{5}{100} = 0.05$ and $n = 100$

$\therefore q = 1 - p = 1 - 0.05 = 0.95$

$\therefore P(x) = {}^nC_x \, p^x \, q^{n-x} = {}^{100}C_x [0.05]^x [0.95]^{100-x}$

The probability of that 2 of these pages contain typographical error $= P[x=2] = {}^{100}C_2 [0.05]^2 [0.95]^{98} = 0.081$

(ii) using the poisson distribution

Given that $p = \frac{5}{100} = 0.05$ and $n = 100$

$\therefore$ mean $= m = np = 100 \times 0.05 = 5$

$\therefore P(x) = \frac{e^{-m} m^x}{x!} = \frac{e^{-5} \times 5^x}{x!}$

The probability of that 2 of these pages contain typographical error $= P[x=2] = \frac{e^{-5} \times 5^2}{2!} = 0.084$.

(14). If X is a poisson variate such that $P(X=2) = 9P(X=4) + 90P(X=6)$, find (i) mean and (ii) Standard deviation.

▶ We have $P(x) = \frac{e^{-m} m^x}{x!}$

Given that $P[x=2] = 9P(x=4) + 90P[x=6]$

$$\frac{e^{-m} m^2}{2!} = 9 \frac{e^{-m} \times m^4}{4!} + 90 \frac{e^{-m} m^6}{6!}$$

$$\frac{e^{-m} m^2}{2!} = 9 \frac{e^{-m} \times m^2 \times m^2}{2! \quad 4 \times 3} + 90 \frac{e^{-m} m^2}{2!} \frac{m^4}{6 \times 5 \times 4 \times 3}$$

cancelling $\frac{e^{-m} m^2}{2!}$ from both sides,

$$1 = \frac{9}{12} m^2 + \frac{1}{4} m^4 \Rightarrow m^4 + 3m^2 - 4 = 0$$

$(m^2+4)(m^2-1) = 0$, i.e $m^2 = -4, 1$.

Note that m^2 cannot be -4 and m cannot be -1.

$\therefore m = 1$

Hence (i) mean $= m = 1$ and

(ii) Standard deviation $= \sigma = \sqrt{m} = \sqrt{1} = 1$.